

Self-Calibrated Diffusion Interference Cancellation under Nonstationary Interference

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Abstract—Diffusion interference cancellation relies on an interference scale that is often selected from the true signal-to-interference-plus-noise ratio (SINR). An unknown scale, particularly one that varies within a frame, creates a mismatch between the received signal and the likelihood used for reconstruction. This paper proposes a self-calibrated interference cancellation diffusion model (SC-ICDM) for wireless image transmission. The receiver initializes blockwise interference powers from received energy, then alternates diffusion reconstruction with nonnegative least-squares amplitude updates. The updates also adapt the local likelihood guidance and reuse existing denoiser outputs without retraining. An exploratory evaluation over 200 image pairs, three interference profiles, and six SINR values shows a mean PSNR gain of 0.599 dB over one-shot blockwise calibration, with a 95% image-pair-cluster bootstrap interval of 0.551–0.648 dB. The gain reaches 1.478 dB relative to the true-global-SINR reference. However, updates degrade reconstruction under severe burst interference, and direct decoding remains preferable at the weakest tested interference. These results establish the value of local receiver calibration in the tested setting while identifying its operating limits.

Index Terms—semantic communications, interference cancellation, diffusion models, SINR estimation, nonstationary interference

I. INTRODUCTION

A wireless image receiver must reconstruct its source from channel observations that can contain both noise and structured interference. Deep joint source-channel coding (JSCC) learns this mapping through an encoder and decoder trained across the channel [1]. Semantic communication methods can additionally shape the representation to preserve information needed for downstream tasks, as demonstrated by contrastive learning for image transmission [2]. At the receiver, however, an interfering transmission need not resemble the Gaussian perturbations used during training. Its structure and power then become relevant to reconstruction.

Learned interference cancellation offers two approaches to this problem. A direct separation network maps the received mixture to the desired waveform; for example, Fourier-convolution networks have been studied for blind co-channel cancellation [3]. A generative receiver instead uses learned priors to constrain the components that explain the observation. Channel denoising diffusion models (CDDMs) apply such a prior to suppress channel noise [4]. The interference cancellation diffusion model (ICDM) extends this principle by coupling separate desired signal and interference priors through a shared

likelihood [5]. The latter approach is attractive when pretrained priors are available and adaptation can be confined to inference.

The ICDM implementation considered here selects its interference amplitude and guidance coefficients from the true frame-level SINR. Two difficulties arise when that information is unavailable. First, the receiver must infer the scale from the mixture itself. Second, even an accurate frame average cannot describe a transmission in which interference power changes across blocks. A global energy estimate addresses the missing side information, but discards these local differences. Independent block estimates preserve them at the cost of greater sensitivity to local signal energy and finite-sample cross terms.

The diffusion trajectory provides another source of information. As reconstruction proceeds, the receiver obtains estimates of both components of the mixture. Given those estimates, fitting a nonnegative interference amplitude requires only a scalar least-squares problem per block. The fitted amplitudes can then be returned to the likelihood correction. This alternation uses information already computed by the receiver and keeps the pretrained networks fixed. Joint estimation of signals and unknown forward parameters has precedents in blind diffusion inverse problems, including GibbsDDRM [6]; the question here is whether a simple blockwise point update is useful within an ICDM communication receiver.

We propose SC-ICDM, a receiver that initializes its local powers from received energy and refines them during reverse diffusion. Our contributions are threefold:

- A SINR-blind formulation of ICDM with blockwise interference amplitudes and receiver information specified explicitly.
- A closed-form calibration update that adapts both the interference scale and local guidance without additional denoiser evaluations.
- A matched evaluation separating global estimation, local initialization, and iterative refinement, including the regimes where refinement or cancellation reduces reconstruction quality.

Across the tested profiles, SC-ICDM improves mean PSNR by 0.599 dB over one-shot blockwise calibration. The result concerns reconstruction: the estimated powers do not consistently approach the physical powers. Section II defines the signal model and receiver assumptions. Section III develops the calibration procedure, and Section IV evaluates its gains and limitations before Section V concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Image Transmission with Blockwise Interference

Let s denote the source image and let $\mathbf{x} = f_{\text{enc}}(s) \in \mathbb{C}^N$ be its transmitted JSCC representation, normalized to $\|\mathbf{x}\|_2^2/N = 1$. An independent interference image produces a latent vector $\mathbf{z} \in \mathbb{C}^N$ with the same framewise normalization. A fixed partition $\{\mathcal{B}_b\}_{b=1}^B$ divides the transmitted symbol order into blocks of length $L_b = |\mathcal{B}_b|$. We consider the AWGN observation

$$\mathbf{y}_b = \mathbf{x}_b + a_b \mathbf{z}_b + \mathbf{n}_b, \quad \mathbf{n}_b \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}), \quad (1)$$

where $a_b \geq 0$ is constant within block b but unknown to the receiver. One amplitude multiplies both components of each complex symbol. The nominal interference power and nominal SINR are

$$\bar{p} = \frac{1}{N} \sum_{b=1}^B L_b a_b^2, \quad \gamma = -10 \log_{10}(\bar{p} + N_0). \quad (2)$$

Here a_b^2 is a model coefficient. The realized interference energy in block b is $a_b^2 \|\mathbf{z}_b\|_2^2 / L_b$, which need not equal a_b^2 because normalization is applied to the complete frame.

The receiver returns a desired latent estimate $\hat{\mathbf{x}}$ and reconstructs the image as $\hat{s} = f_{\text{dec}}(\hat{\mathbf{x}})$. We hold the encoder, decoder, and diffusion priors fixed so that changes in reconstruction can be attributed to the receiver procedure.

B. Available Information and Calibration Objective

The receiver observes $\mathbf{y}$ and knows N_0 , the block partition, and the pretrained desired and interference priors. It has no access to the clean latents, block amplitudes, profile label, or true SINR. Thus, *SINR-blind* refers specifically to the missing interference-power information; the noise level and interference family remain known.

For fixed $\mathbf{a} = (a_1, \dots, a_B)$, the Gaussian model gives

$$-\log p(\mathbf{y} | \mathbf{x}, \mathbf{z}, \mathbf{a}) = \frac{1}{N_0} \sum_b \|\mathbf{y}_b - \mathbf{x}_b - a_b \mathbf{z}_b\|_2^2 + C, \quad (3)$$

where C is independent of the latent vectors and amplitudes. ICDM uses its two priors to guide recovery of these latent vectors. Our task is to supply useful local amplitudes throughout that recovery when $\mathbf{a}$ is unknown. The objective is image reconstruction; accurate physical power estimation is a separate property to be tested.

III. SELF-CALIBRATED DIFFUSION RECEIVER

A. Initialization from Received Energy

Under unit expected signal and interference power and vanishing cross moments, the expected received energy in a block is $1 + a_b^2 + N_0$. This motivates the initialization

$$\hat{p}_b^{(0)} = \left[\|\mathbf{y}_b\|_2^2 / L_b - 1 - N_0 \right]_+, \quad \hat{a}_b^{(0)} = \sqrt{\hat{p}_b^{(0)}}, \quad (4)$$

where $[u]_+ = \max(u, 0)$. Using the full frame in (4) gives a global estimate; using individual blocks retains local variation. The latter is an energy proxy, since framewise normalization

does not enforce unit power in each block. Local latent energy, noise fluctuations, and signal–interference cross terms all contribute to its error.

For any current power estimate, the corresponding local SINR is

$$\hat{\gamma}_b = -10 \log_{10}(\hat{p}_b + N_0). \quad (5)$$

We obtain the guidance coefficients (λ_b, β_b) by linear interpolation of the fixed ICDM lookup table. At the AWGN SINR knots $(-7, -4, 0, 4, 7, 10, 20)$ dB, the respective coefficient sequences are $(0.3, 1, 1.4, 3, 4.5, 5.5, 3)$ and $(0.1, 0.7, 1, 1, 1, 0.5, 2)$. Values outside the range use the nearest endpoint. These inherited coefficients are not optimized on the reported image pairs.

B. Amplitude Refinement from Denoised Estimates

Let α_t and σ_t denote the forward diffusion signal and noise coefficients. At a reverse step, the two frozen noise predictors provide clean-point estimates

$$\begin{aligned} \hat{\mathbf{x}}_0^{(t)} &= \frac{\mathbf{x}_t - \sigma_t \epsilon_{\theta_x}(\mathbf{x}_t, t)}{\alpha_t}, \\ \hat{\mathbf{z}}_0^{(t)} &= \frac{\mathbf{z}_t - \sigma_t \epsilon_{\theta_z}(\mathbf{z}_t, t)}{\alpha_t}. \end{aligned} \quad (6)$$

Before fitting amplitudes, we normalize each complete-frame estimate:

$$\Pi(\mathbf{u}) = \frac{\mathbf{u}}{\sqrt{\max(\|\mathbf{u}\|_2^2 / N, \epsilon_{\text{mach}})}}, \quad \tilde{\mathbf{x}}_0^{(t)} = \Pi(\hat{\mathbf{x}}_0^{(t)}), \quad (7)$$

and similarly for $\tilde{\mathbf{z}}_0^{(t)}$. This projection imposes the transmitter power convention on the estimated components. It is applied over the frame rather than separately within each block, allowing local latent-energy variation to remain.

For fixed projected components, the likelihood in (3) separates into independent scalar problems:

$$\min_{a \geq 0} \left\| \mathbf{y}_b - \tilde{\mathbf{x}}_{0,b}^{(t)} - a \tilde{\mathbf{z}}_{0,b}^{(t)} \right\|_2^2. \quad (8)$$

Define $h_b = \left\| \tilde{\mathbf{z}}_{0,b}^{(t)} \right\|_2^2$ and $c_b = \text{Re}\{(\tilde{\mathbf{z}}_{0,b}^{(t)})^H (\mathbf{y}_b - \tilde{\mathbf{x}}_{0,b}^{(t)})\}$. The objective is $h_b a^2 - 2c_b a + \text{const}$. Differentiating and imposing nonnegativity gives the implemented update

$$\hat{a}_b^{\text{new}} = \begin{cases} [c_b/h_b]_+, & h_b > L_b \epsilon_{\text{mach}}, \\ \hat{a}_b^{\text{old}}, & \text{otherwise.} \end{cases} \quad (9)$$

The fallback prevents division by a numerically empty estimate. There is no ridge penalty or amplitude cap in the evaluated update.

Calibration begins when $\alpha_t \geq 0.5$; earlier steps retain the energy initialization. Because (6) divides by α_t , this delay avoids fitting from clean-point estimates at the noisiest end of the trajectory. It is a fixed activation rule, not an estimate of whether a particular block is reliable. The evaluated 40-step, order-3 UniPC schedule performs 21 calibration updates.

C. Feedback into the ICDM Sampler

After each amplitude update, the receiver sets $\hat{p}_b = (\hat{a}_b^{\text{new}})^2$ and recomputes the local guidance coefficients using (5). To make the role of calibration explicit, the AWGN likelihood correction inherited from ICDM can be written, at the states entering that correction, as

$$\mathbf{r}_{t,b} = \mathbf{y}_b / \zeta_t - \mathbf{x}_{t,b} - \hat{a}_b \mathbf{z}_{t,b}, \quad \zeta_t = (2 - \alpha_t^2)^{-1}, \quad (10)$$

$$\begin{aligned} \epsilon_{x,b}^g &= \epsilon_{x,b} - \zeta_t^2 \lambda_b \mathbf{r}_{t,b}, \\ \epsilon_{z,b}^g &= \epsilon_{z,b} - \zeta_t^2 \beta_b \hat{a}_b \mathbf{r}_{t,b}. \end{aligned} \quad (11)$$

Here $\epsilon_{x,b}$ and $\epsilon_{z,b}$ are the noise predictions used by the sampler. The block amplitudes and coefficients are broadcast over their associated symbols. The numerical integration and likelihood correction otherwise follow the retained ICDM implementation.

Equations (9)–(11) form the feedback loop: the current reconstruction determines the next amplitude, and that amplitude changes the subsequent reconstruction. The update changes both the mixture scale and the guidance strength. Accordingly, the experiments evaluate their combined effect; they do not isolate either factor.

The complete procedure is summarized below. It reuses the two noise predictions already required at each eligible corrector step.

SC-ICDM inference

- 1: Initialize $\hat{p}_b, \hat{a}_b$ by (4).
- 2: Initialize the two diffusion states as in ICDM.
- 3: **for** each reverse sampling step **do**
- 4: Advance the two UniPC predictors.
- 5: **if** a corrector evaluation is available **then**
- 6: Evaluate the two frozen noise predictors.
- 7: **if** $\alpha_t \geq 0.5$ **then**
- 8: Form and project the estimates in (6)–(7).
- 9: Update all amplitudes by (9).
- 10: **end if**
- 11: Apply the UniPC corrector and local guidance using the current amplitudes and coefficients.
- 12: **end if**
- 13: **end for**
- 14: Normalize the recovered desired latent and decode.

Each calibration pass requires linear work in N for normalization, inner products, and blockwise reductions, plus $O(B)$ coefficient interpolation. It adds no denoiser call or trainable parameter. The least-squares step minimizes (8) for fixed point estimates, but those estimates change during diffusion. Consequently, this local minimization does not establish monotonic improvement of image quality or convergence of a joint posterior objective.

IV. SIMULATION RESULTS

A. Experimental Settings

We evaluate 200 CelebA desired images paired with CIFAR-10 interference images, reconstructing the desired images

at 128×128 resolution. All receivers use the same frozen SwinJSCC encoder and decoder [7], interference encoder, and two DiT diffusion priors. The latent bottleneck has eight channels. Noise SNR is fixed at 20 dB; nominal SINR takes values in $\{-7, -4, 0, 4, 7, 10\}$ dB. Each pair is evaluated under three channel seeds and three interference profiles, giving 10,800 reconstructions per receiver.

Blocks contain 64 complex symbols. The stationary profile has equal block powers; alternating and burst profiles repeat power multipliers (0.25, 1.75) and (0, 2), respectively. The multipliers are normalized to preserve the nominal frame average. These profiles test the effect of local power variation while keeping the nominal SINR fixed. Their boundaries coincide with the receiver partition.

We compare the following receivers:

- *Direct*: JSCC decoding without diffusion cancellation.
- *Global-SINR* (A0): ICDM with true global SINR and the original amplitude convention.
- *Global-blind* (A2): one received-energy estimate for the complete frame.
- *Block-blind* (A3): blockwise initialization from (4), held fixed throughout diffusion.
- *SC-ICDM* (A4): the proposed alternating calibration.

Global-SINR knows the nominal frame average but not the local profile, so it is a reference rather than an upper bound. An additional control (A1) corrects its amplitude convention while keeping the true SINR. Every diffusion receiver uses 40 order-3 UniPC steps with the same number of network evaluations.

Comparisons are paired by desired image, interference image, channel seed, and diffusion initialization. PSNR is computed per reconstruction before averaging. For SC-ICDM contrasts, we report 95% percentile bootstrap intervals from 10,000 resamples of the 200 image-pair clusters, retaining all seeds and operating conditions within each selected cluster. The image pairs were inspected during method development. The intervals therefore describe exploratory results and are not adjusted for multiple comparisons.

B. Global and Blockwise Calibration

Table I separates the effect of estimating the global power from that of retaining local variation. Global-blind differs from Global-SINR by only -0.005 , -0.008 , and -0.005 dB for stationary, alternating, and burst interference, respectively. Thus, a blind energy estimate closely reproduces the mean reconstruction quality of the global reference in this experiment. These small descriptive differences do not by themselves establish statistical equivalence. Fig. 1 gives the corresponding PSNR curves, retaining the individual SINR values that are averaged in the table.

Block-blind gains 0.819 dB and 2.108 dB over Global-SINR in the alternating and burst cases. It loses 0.292 dB under stationary interference, where separate blocks contain no additional power-profile information. The pattern is consistent with the trade-off in (4): shorter averaging windows retain local changes but also retain local fluctuations. Correcting only

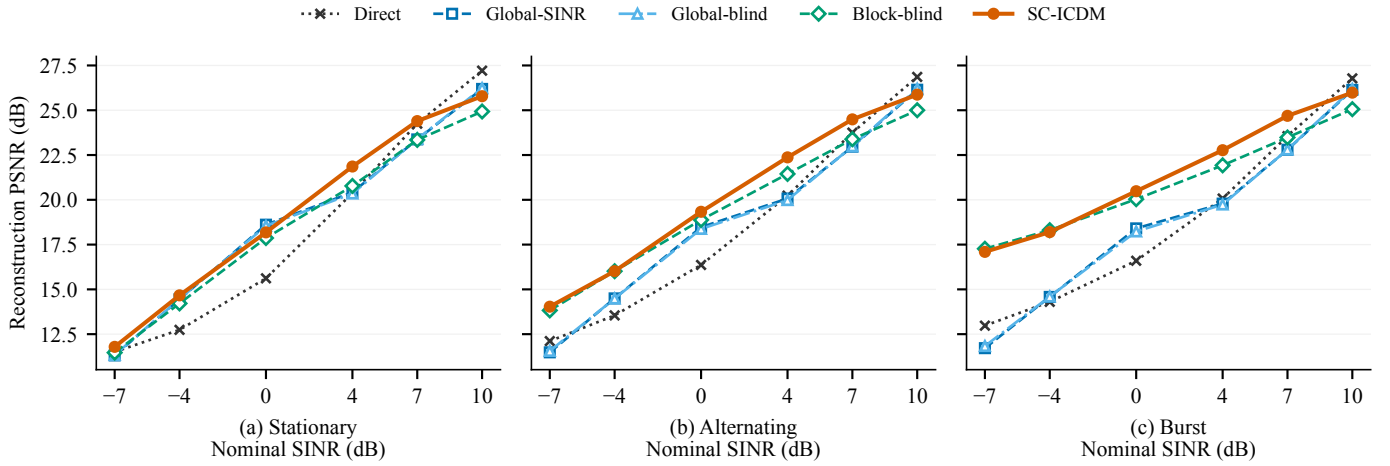


Fig. 1. Reconstruction PSNR versus nominal SINR under the three power profiles. Each point averages 200 image pairs and three channel seeds. Global-SINR and Global-blind nearly overlap. The direct-decoding curves show why an improvement over another cancellation receiver need not imply an improvement over omitting cancellation.

TABLE I
MEAN PSNR (dB) OVER 200 IMAGE PAIRS, THREE SEEDS, AND SIX SINR VALUES. THE FINAL COLUMN ALSO AVERAGES OVER THE THREE PROFILES.

Receiver	Stationary	Alternating	Burst	All
Direct	18.616	18.811	19.047	18.825
Global-SINR	19.058	18.939	18.903	18.967
Global-blind	19.053	18.931	18.898	18.961
Block-blind	18.767	19.758	21.012	19.845
SC-ICDM	19.446	20.352	21.535	20.444

the original amplitude convention changes mean PSNR by 0.007–0.010 dB across profiles, far less than the nonstationary blockwise gains.

C. Benefit and Limits of Iterative Refinement

SC-ICDM improves Block-blind by 0.599 dB overall, with a 95% cluster interval of 0.551–0.648 dB. The profile-specific gains are 0.680 dB for stationary interference, 0.594 dB for alternating interference, and 0.523 dB for burst interference, with respective intervals of 0.629–0.729, 0.540–0.649, and 0.469–0.578 dB. The proposed receiver exceeds Block-blind in 79.1% of matched trials. Relative to Global-SINR, the overall gain is 1.478 dB (1.392–1.563 dB).

Fig. 2 shows how refinement depends on the operating point. Averaging across profiles, the gain over Block-blind increases from 0.118 dB at –7 dB SINR to 1.122 dB at 7 dB SINR. The burst profile is the exception at strong interference: refinement loses 0.174 dB at –7 dB SINR and 0.109 dB at –4 dB SINR. The corresponding intervals for the signed differences are [–0.224, –0.126] and [–0.197, –0.025] dB. For alternating interference at –4 dB, the difference is just 0.001 dB with interval [–0.092, 0.092] dB. An average improvement therefore does not justify updating every interference realization.

The comparison with Direct answers a different question: whether cancellation itself is beneficial. At 10 dB SINR, SC-ICDM gains 0.882 dB over Block-blind but is still 1.070 dB

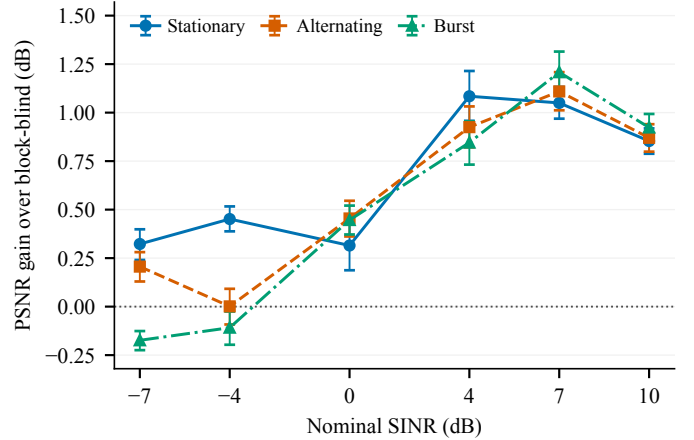


Fig. 2. Paired PSNR gain of SC-ICDM over Block-blind. Bars show exploratory 95% image-pair-cluster bootstrap intervals. Negative values at low SINR under burst interference identify operating points where refinement hurts.

below Direct (interval [–1.270, –0.867] dB). At this operating point, improved calibration does not remove the disadvantage of invoking the diffusion receiver. A practical extension is to choose between direct decoding and cancellation from receiver-observable evidence; no such selection rule is evaluated here.

D. Power Estimates and Computational Cost

The final amplitudes should not be interpreted as consistently improved physical-power estimates. Although SC-ICDM improves reconstruction in 79.1% of trials, its block-power RMSE improves over initialization in only 28.3%. At SINR values of 0, 4, and 10 dB, median final power estimates are 0.234, 0.024, and 0.004, compared with nominal frame powers of 0.990, 0.388, and 0.090. Fourteen trials contain an estimated block power above 100; the maximum is 93,186.656. The numerical fallback in (9) therefore does not prevent all large estimates.

These diagnostics support interpreting the update as a reconstruction calibration mechanism. They do not support a claim of accurate power recovery. Small interference-estimate energy can amplify the ratio in (9), but the recorded outputs do not establish which mechanism causes the observed outliers or the severe-burst losses. Separate scale and guidance ablations would also be needed to determine which part of the feedback contributes most to the PSNR gain.

Measured receiver time is 0.323 s per image for Block-blind and 0.347 s for SC-ICDM on an RTX 4090 with batch size four. Timing includes calibration, sampling, output normalization, and decoding, with GPU synchronization; encoding, data loading, and channel generation are excluded. The measurements come from separate runs and should be read as descriptive timings, not as a controlled estimate of overhead or batch-one latency.

The evaluation covers AWGN with known noise power and block boundaries, one desired/interference dataset pairing, one block length, and matched frozen priors. It does not establish performance under fading, unknown change points, or prior mismatch. PSNR is the only recorded reconstruction metric, and independent image pairs are needed to confirm the gains after method selection. These limits define the evidence available for the receiver rather than assumptions about its performance in untested settings.

V. CONCLUSION

We proposed a self-calibrated diffusion receiver for image transmission with unknown blockwise interference power. Received-energy initialization removes the true-SINR input, and closed-form amplitude updates feed the evolving diffusion estimates back into local guidance. In the exploratory

evaluation, this feedback improves mean PSNR by 0.599 dB over one-shot blockwise calibration without additional denoiser evaluations. The benefit is conditional: refinement loses under severe burst interference, and direct decoding wins at the weakest tested interference. The results motivate reliability-based update and bypass rules, together with independent validation and perceptual evaluation.

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